DATAX500

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Github Repository
Moodle Submission
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Assignment 1

1. Using the '*heart.csv*' data-set, calculate the following statistical measures for age in *R*

First, we load the data-set:

```
# load automatically via git files
heart<-read.csv("Heart.csv")


## Alternatively, load the dataset manually using:
# heart<-read.csv(file.choose())
```

(6 marks)

a) Mean

```
# assign age column to "age" var for easier processing
age<-heart$age

# calculate the mean age
mean_age<-mean(age)
mean_age
```

```
## [1] 54.01038
```

```
# round to 2 d.p.
mean_age<-round(mean_age, 2)
mean_age
```

```
## [1] 54.01
```

(2 marks)

b) Median

```r
# calculate the median age
median_age<-median(age)
median_age
```

```
## [1] 54
```

(2 marks)

c) Sample standard deviation

```r
# calculate the sample standard deviation
sd_age<-sd(age)
sd_age
```

```
## [1] 9.132316
```

```r
# round sample standard deviation to 2 d.p.
sd_age<-round(sd_age, 2)
sd_age
```

```
## [1] 9.13
```

(2 marks)

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2. A researcher wishes to investigate the relationship between *gender* and *cholesterol levels*

Key-points to notice here is that **we are measuring cholesterol levels** based on gender, making **cholesterol** the *dependent variable*, and **gender** the *independent variable*.

Using the heart.csv data set, select and perform an *appropriate statistical test* in *R* and interpret the output. Your answer should include a **null hypothesis**, an **alternative hypothesis** , and a **p-value**

Since *gender* (categorical) and *cholesterol* (continuous) are **independent** from each other, we can conduct an independent (two-sampled) t-test to test the following hypotheses:

Null hypothesis:

- There will be no difference in mean cholesterol levels between genders. (H0: U1 = U2)

Alternative hypothesis:

- There will be a significant difference in mean cholesterol levels between genders. (H1: U1 != U2)

Results:

```r
# peek at the dataset so we can view the column names
head(heart, n=2)
```

```
##   age gender cp trtbps chol fbs restecg maxheartrate exng
## 1  60      1  3    145  233   1       0          150    0
## 2  35      1  2    130  250   0       1          187    0
```

```r
# perform a t-test on the heart data set between chol and gender
t.test(chol~gender, data=heart)
```

```
##
##    Welch Two Sample t-test
##
## data:  chol by gender
## t = 2.9765, df = 133.12, p-value = 0.003464
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##   7.289758 36.169316
## sample estimates:
## mean in group 0 mean in group 1
##        262.6989        240.9694
```

***p*-value:**

$p = 0.003$ (3.d.p.)

Interpretation:

- The Welch two-sample t-test provides **evidence to reject the null hypothesis**, that **there will be no difference in mean cholesterol levels between genders, *p*=0.003** (3 d.p.).
- This interpretation is further supported with **a large t-value of 2.98.**
- **Further testing is recommended** to determine the source or magnitude of this cholesterol difference.
- With *95% confidence*, we can assume that the **true mean difference** will be somewhere between 7.29 and 36.17 (2 d.p.)**.
- Broadly analyzing the data, an apparent **mean difference of ~22** between group 0 and group 1 exists, providing more **evidence to reject the null** hypothesis, that there is no difference between the means, regardless of gender (for each group).

(5 marks)

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3. A researcher wishes to investigate the predictors of maximum heart rate. Using the data set provided, fit an appropriate statistical model of your choice. Make sure to include age as one of the predictors

a) What are the null and alternative (two-sided) hypotheses associated with age?

- **Null hypothesis:** H0: B1 = 0
  - The regression coefficient for age is zero (i.e., age has no linear effect on maximum heart rate.)

  Note: *This means that for every* x *increase in age, no predictable change in* y *is observed.*

- **Alternative hypothesis:** H1: B1 != 0
  There will be a significant relationship between age and maximum heart-rate.

(2 marks)

b) What is the estimated coefficient of age and what does this imply in terms of the relationship between age and maximum heart rate?

```r
# again, peek at the dataset so we can see what we're working with
head(heart, n=2)
```

```
##   age gender cp trtbps chol fbs restecg maxheartrate exng
## 1  60      1  3    145  233   1       0          150    0
## 2  35      1  2    130  250   0       1          187    0
```

```r
# store the mean maxheartrate in a variable
rate<-heart$maxheartrate

# create a multiple linear regression model to predict max heart rate
# given an age and gender
model<-lm(rate ~ age + gender, data=heart)

# ouput the model summary for analysis
summary(model)
```

```
##
## Call:
## lm(formula = rate ~ age + gender, data = heart)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -64.252 -11.707   3.278  15.444  46.388
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 209.1728     7.8858  26.525  < 2e-16 ***
## age          -1.0277     0.1363  -7.542 6.17e-13 ***
## gender       -5.0680     2.6589  -1.906   0.0577 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.96 on 286 degrees of freedom
## Multiple R-squared:  0.1683,   Adjusted R-squared:  0.1624
## F-statistic: 28.93 on 2 and 286 DF,  p-value: 3.612e-12
```

Estimated coefficient of age:

-1.0277

Implications (in terms of relationship with max heart rate):

This model indicates a *negative linear* relationship. For *every one-year increase in* ***age****, the* ***maximum heart rate*** *is estimated to* ***decrease by ~1.028 bpm (3 d.p.)***, regardless of gender.

(3 marks)

c) Comment on the predictors you have included in the model and justify the inclusion of these variables in your model

I chose to use **age** and **gender** for my predictors in this model. These are both *common demographic predictors* and often enable clearer, more informative interpretations.

It is understood that ones cardiovascular health typically deteriorates with age, but it's unclear if this is true for both Men, and Women. Using **age** and **gender** as *predictors* in this model will allow us to estimate the relationship between **age** and **maximum heart rate** while **controlling for gender**

- **Age:** *Age* is a **continuous** variable, making it **suitable for regression**. I chose this not only due to its common use in other studies, especially health studies, but also because an individual's maximum heart rate is likely to decline as they get older - this predictor allows us to quantify that change.
- **Gender:** *Gender* is a categorical variable (0=male, 1=female). Controlling for gender helps to distinguish differences between Males and Females and is likely to increase the accuracy of our estimations.

(3 marks)

d) Interpret the model output in plain English. Your interpretation must include the p-value associated with each coefficient (slopes and intercept) test, and R-squared value

The model provides the following insights:

- Age is a highly significant predictor (p = 6.17e-13, t = -7.542). We can say with 99.9% confidence that as age increases, maximum heart rate decreases, regardless of gender.
- Gender (p = 0.058 (3.d.p.)) is slightly above the the 0.05 threshold, suggesting that while there is a trend, gender is not a statistically significant predictor in this model, at least, not at the 95% confidence level.
- Intercept (p < 2e-16) is highly significant, much lower than the threshold of 0.05. This suggests that the baseline of this model is statistically valid.

- Model accuracy: The multiple R squared value of 0.1683 suggests that only approximately 16.8% of the variance in our data can be explained by age and gender. While some relationship exists, and is significant, there are probably some other factors that could better predict the change in maximum heart rate as age increases.

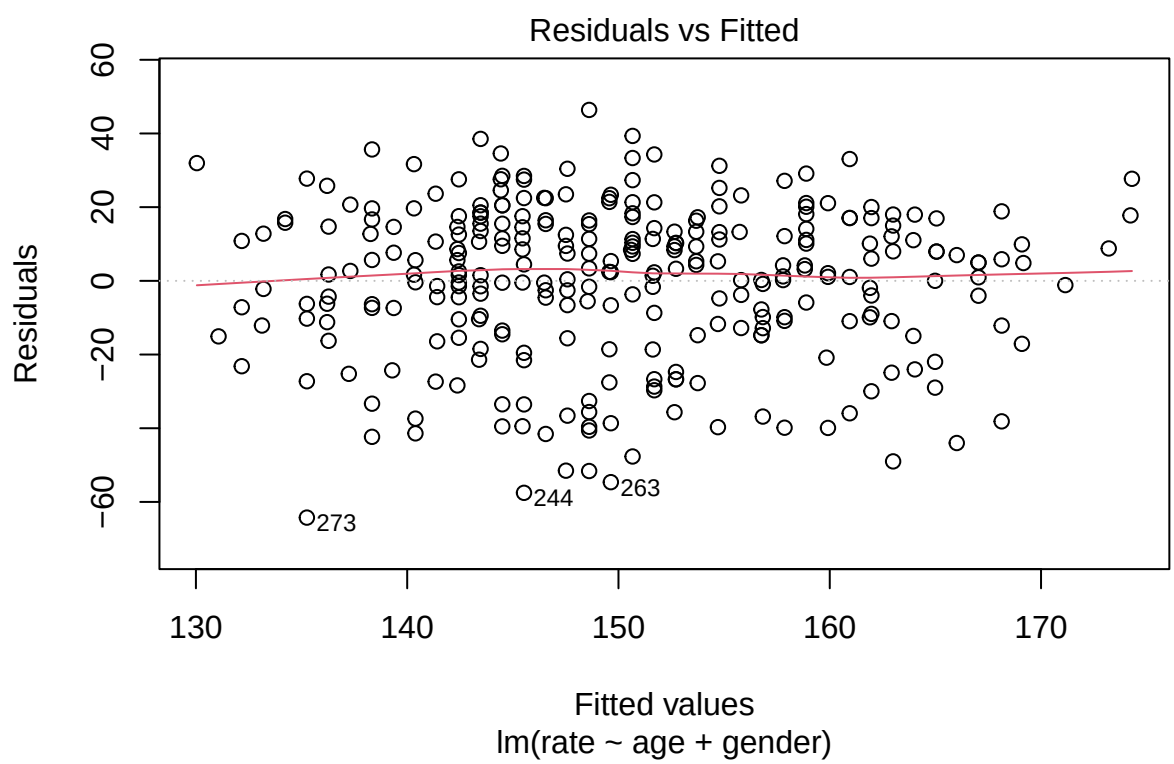
In summary, we have strong evidence to reject the null hypothesis for age, that the regression coefficient will be zero, regardless of gender.

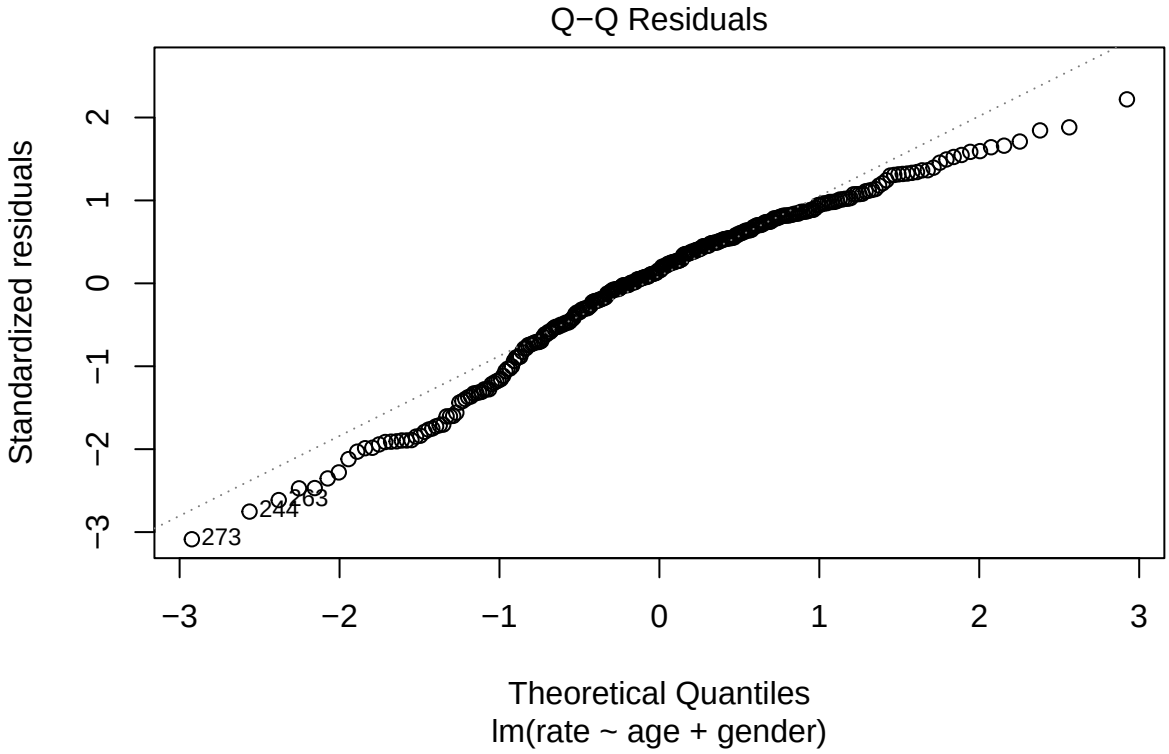
(3 marks)

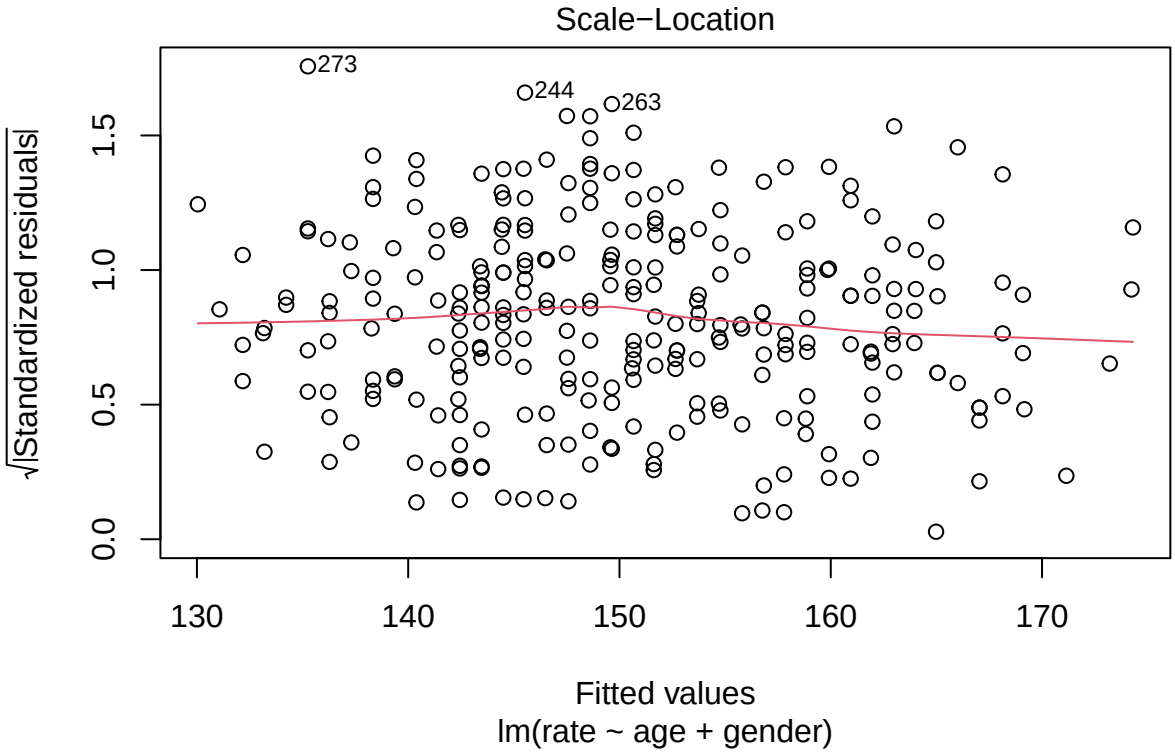
e) Comment on whether the assumptions of a linear regression model have been met. Include appropriate statistical plots to support your answer

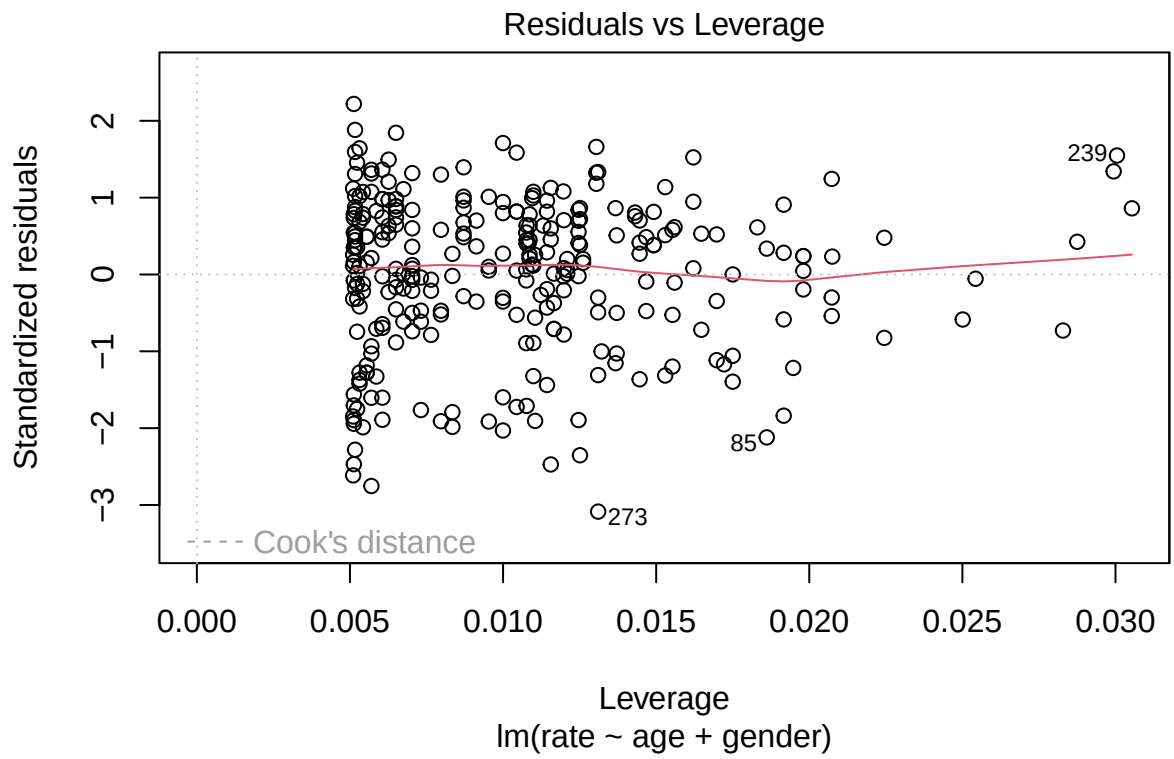
Appropriate statistical plots:

```
# generate some plots to evaluate the regression assumptions:
  ## Linearity, Independence, Homoscedasticity and Normality

# create a 2x2 grid of diagnostic plots
#par(mfrow=c(2,2)) # uncomment for small grid-style output

# plot appropriate statistical plots
plot(model)
```




Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(rate ~ age + gender)
273
244
263
−3
−2
−1
0
1
2
3


Scale−Location
273
244
263
√|Standardized residuals|
0.0
0.5
1.0
1.5
130
140
150
160
170
Fitted values
lm(rate ~ age + gender)



e) Commentary on Model Assumptions

Based on the diagnostic plots, the assumptions for this linear regression are largely satisfied, though there are a few minor deviations:

Linearity (Residuals vs Fitted):

The red line is relatively flat and stays close to the zero line. This suggests that the relationship between the predictors (age and gender) and maximum heart rate is correctly modeled as linear. There is no obvious "U" shape that would suggest we missed a non-linear trend.

###Normality (Q-Q Residuals):

The points follow the dashed diagonal line very closely through the middle. There is a tiny bit of "wavering" at the extreme ends (specifically points 273 and 239), but it's not severe enough to violate the assumption of normally distributed residuals.

Homoscedasticity (Scale-Location):

The red line is nearly horizontal, and the "cloud" of points has a fairly consistent vertical spread across the whole x-axis. This indicates constant variance (no "fan" shape), meaning the model is equally accurate (or inaccurate) across all age groups.

Influential Outliers (Residuals vs Leverage):

All points fall well within the dashed red boundaries for Cook's Distance. While points 273, 239, and 85 are labeled as having higher leverage or larger residuals, they aren't "pulling" the model enough to be considered problematic outliers that would invalidate the results.

Independence (Study design):

Since each observation in this data set represents a separate, unrelated individual patient with no repeated measures, independence can be assumed.

(3 marks)

(14 marks)

Data dictionary:

cp: chest pain type

1. typical angina
2. atypical angina
3. non-anginal pain
4. asymptomatic

trtbps:

- resting blood pressure (in mm Hg)

chol:

- cholesterol (in mg/dl)

gender:

0. male;
1. female;

fbs: (fasting blood sugar > 120 mg/dl)

0. False
1. True

rest_ecg: resting electrocardiographic result

0. Normal
1. Having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV)
2. Showing probably or definite left ventricular hypertrophy by Estes' criterion

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Regression Assumption Notes:

Linearity:

- **Residual/fitted plots** should be relatively flat and close to zero.
- A curved or "U-shaped" line suggests a non-linear relationship (consider transforming or adding polynomial terms)

Normality:

- The **Q-Q Residuals plots** should follow the diagonal dashed line closely.
- Minor deviation at the tails is acceptable, but strong "S-curves" or heavy tailing suggests non-normal residuals (consider transforming the response variable)

Homoscedasticity:

- The **Scale-Location plots'** red line should be nearly horizontal with an even spread of points. A "fan" shape (spread increasing left to right) indicates heteroscedasticity (constant variances is violated, consider transformation or weighted regression)

Independence:

- Cannot be visually verified, but rather, justified by study design. If each observation is a separate, unrelated individual/event with no repeated measures or time-series structure, independence can be reasonably assumed.

Influential Outliers:

- Any points outside Cook's Distance boundaries (dashed red lines) in the **Residuals vs. Leverage plot** are influential, and may be distorting the model (consider investigating or removing them with justification)